

To find the derivative of $\int_0^T c(\beta(t))dt$ with respect to β we use that (assuming $c(\cdot)$ is of C^1 class) for any variation $\delta\beta \in L^2[0, T]$:

$$\int_0^T c(\beta(t) + \delta\beta(t))dt = \int_0^T c(\beta(t))dt + \langle c'(\beta(\cdot)), \delta\beta(\cdot) \rangle_{L^2[0, T]} + o(\|\delta\beta\|_{L^2[0, T]}). \quad (3.30)$$

This relation can be proved easily using the Taylor expansion for c ; by the definition of the Fréchet derivative in $L^2[0, T]$ (see section 3.6.2) we obtain thus :

$$\frac{\partial}{\partial \beta} \left[\int_0^T c(\beta(t))dt \right] = c'(\beta(\cdot)). \quad (3.31)$$

On the other hand, the epidemic size $S(0) - S(T)$ is less straightforward to differentiate because β does not appear explicitly, although it is involved in this function.

The methodology presented in 3.17 can be applied ; we introduce the adjoint states λ and μ solution of :

$$\lambda'(t) = \beta(t)I(t)(\lambda(t) - \mu(t))/N_p \quad (3.32)$$

$$\mu'(t) = \beta(t)S(t)(\lambda(t) - \mu(t))/N_p + \gamma\mu(t) \quad (3.33)$$

$$(\lambda, \mu)(T) = (-1, 0). \quad (3.34)$$

Then $\frac{\partial}{\partial \beta} S(T) = -SI(\lambda - \mu)/N_p$ (the equality holds as functions of t).

3.6 Theoretical appendix

3.6.1 Graph Theory

We will introduce some notions of graph theory.

Definition 3.18 (Graph). A (directed /oriented) graph is a pair $G = (V, E)$ consisting of:

- a set of vertices V ;
- a set of edges $E \subseteq V \times V$, with $(a, b) \in E$ meaning that there is an edge between a and b . If the graph is oriented then the edge is **from** a **to** b and in this case the edge may be called an “arc”.

Example 3.19. See figure 3.9 for an example of a directed graph.

If the graph is oriented, the edge (a, b) is not the same as (b, a) (if $a \neq b$); note the it is possible for an edge/arc to point to the node from which it originates i.e. (a, a) .

Remark 3.20. An undirected graph is a graph $G = (V, E)$ such that for any $(x, y) \in E$ we also have $(y, x) \in E$.

Some other definitions will be useful in the following.

Definition 3.21.

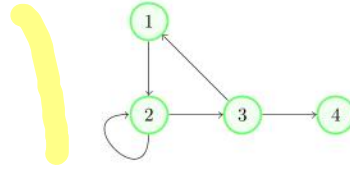


Figure 3.9: Example of a oriented graph. Here $V = \{1, 2, 3, 4\}$, $E = \{(1, 2), (3, 1), (2, 2), (2, 3), (3, 4)\}$.

1. In a directed or undirected graph, a path from node 'a' to node 'b' is a consecutive sequence of edges connecting the two nodes.
2. A cycle is a path where the first node and the last one are the same;
3. a graph without cycles is called 'acyclic'
4. A graph is said to be connected if it for any pair of nodes $v_1, v_2 \in V$ there exists a path from node v_1 to node v_2 .
5. in a directed grapg G the in-neighborhood of a vertex $b \in V$ is the set of all precedents of b :

$$N_G^-(b) = \{a \in V, (a, b) \in E\}. \quad (3.35)$$

Example 3.22. See figure 3.10 for examples of connected or not connected graphs.

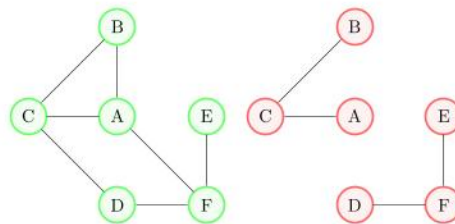


Figure 3.10: **Left :** connected graph. **Right :** non-connected graph

The graph in Figure 3.10 (left) has several cycles (notably $\{A, F, D, C\}$), while the one on the right has none.

Definition 3.23 (Node Degrees). Given an arc (x, y) connecting nodes x and y , we call the starting node x the **source** and the ending node y the **target**. We then define:

1. the **out-degree** of a node as the number of arcs that have this node as the source;
2. the **in-degree** of a node as the number of arcs that have this node as the target.

3. A node/vertex 'a' with in-degree null is called a 'source' or 'input'; and a vertex 'a' null out-degree is called a 'sink' or 'output' node.

Note: a graph thus has as many in-degrees as out-degrees.

Example 3.24. In the graph of Figure 3.9:

- the out-degree of node {2} is 2, the out-degree of node {4} is zero.
- the in-degree of node {3} is 1, the in-degree of node {2} is 2.
- There are a total of 5 in-degrees and 5 out-degrees, which confirms our definition: there are as many in-degrees as out-degrees.

Definition 3.25. A directed graph $G = (V, E)$ is called a **Directed Acyclic Graph (DAG)** if and only if it satisfies the following conditions:

1. G is a directed graph, meaning that each edge $(u, v) \in E$ has a direction from vertex u to vertex v .
2. G contains no directed cycles, i.e., there does not exist a sequence of vertices $v_1, v_2, \dots, v_k$ such that:

$$(v_1, v_2), (v_2, v_3), \dots, (v_{k-1}, v_k), (v_k, v_1) \in E,$$

where $k \geq 2$.

Proposition 3.26. A **Directed Acyclic Graph** is a directed graph where there is a **topological ordering** of the vertices, i.e.:

$$\text{There exists a mapping } f : V \rightarrow \mathbb{N} \text{ such that } \forall (u, v) \in E : f(u) < f(v). \quad (3.36)$$

Moreover the topological ordering can be taken to be a bijection between V and $\{1, 2, \dots, |V|\}$, where $|V|$ is the cardinality of V .

Proof. $\Rightarrow$ **We prove by induction $|V|$ over that** if G is a DAG there exists a **topological ordering of the vertices of G :**

Of course, when $|V|$ is 0, 1 or 2 the proposition is immediate.

To prove the induction, assume it is true up to $|V|-1$; we first prove that G has at least a source. Suppose on the contrary that there is no source, then for any v its in-degree is at least 1. This would mean that every vertex has at least one incoming edge; iterating backwards and since $|V|$ is finite there has to be a repetition; we obtain thus a cycle contradicting the assumption that G is a DAG. Hence, there exists a source $v^* \in V$. After removing vertex v^* from the graph (along with any edges it carries), the remaining graph G' is still a DAG (since removing a vertex and its outgoing edges doesn't create cycles). By induction hypothesis, a (bijective) topological ordering f' exists for G' and then

$$f : V \rightarrow \mathbb{N}, f(v) = \begin{cases} 0, & \text{if } v = v^* \\ 1 + f'(v) & \text{otherwise} \end{cases} \quad (3.37)$$

is a valid topological ordering for G with all the required properties.

⇐ **If a topological ordering exists, then G is a DAG:**

Assume G has a topological ordering, i.e., there exists a mapping $f : V \rightarrow \mathbb{N}$ such that for every directed edge $(u, v) \in E$, we have $f(u) < f(v)$. Suppose by contradiction that G contains a directed cycle i.e., a sequence of vertices $v_1, v_2, \dots, v_k$ such that $(v_1, v_2), (v_2, v_3), \dots, (v_k, v_1)$ are edges in G . Using the topological ordering we obtain $f(v_1) < f(v_2) < \dots < f(v_k) < f(v_1)$ which leads to the contradiction $f(v_1) < f(v_1)$. $\square$

3.6.2 Fréchet differentiability

Definition 3.27 (Fréchet differentiability). A function $\mathcal{F} : \mathcal{W}_i \rightarrow \mathcal{W}_o$ between two normed spaces $\mathcal{W}_i$ and $\mathcal{W}_o$ is Fréchet differentiable if there exists a bounded linear operator $A : \mathcal{W}_i \rightarrow \mathcal{W}_o$ such that

$$\lim_{\|v\| \rightarrow 0} \frac{\|\mathcal{F}(x+v) - \mathcal{F}(x) - Av\|}{\|v\|} = 0, \quad (3.38)$$

or in equivalent formulation:

$$\mathcal{F}(x+v) = \mathcal{F}(x) + Av + o(\|v\|). \quad (3.39)$$

In this case A is the differential of $\mathcal{F}$ at x and is denoted $D\mathcal{F}(x)$. When $\mathcal{W}_i$ is a Hilbert space and $\mathcal{W}_o = \mathbb{R}$ the Riesz representation theorem implies that $Av = \langle \xi, v \rangle$ for some ξ which is called the gradient or the derivative of $\mathcal{F}$ at x denoted $\nabla \mathcal{F}(x)$ ².

Example 3.28. Consider $x \mapsto \frac{1}{2}\|Wx + b\|^2$ with W a matrix and x and b vectors of compatible dimensions; its Fréchet differential is the application

$$\delta x \mapsto \langle Wx + b, W^T \delta x \rangle \quad (3.40)$$

and its gradient is $W^T(Wx + b)$.

3.6.3 Differentiability of a composed function

Consider now three normed spaces $\mathcal{W}_1, \mathcal{W}_2$ and $\mathcal{W}_3$ and $g : \mathcal{W}_1 \rightarrow \mathcal{W}_2, f : \mathcal{W}_2 \rightarrow \mathcal{W}_3$ and $F = f \circ g : \mathcal{W}_1 \rightarrow \mathcal{W}_3$. Let $x \in \mathcal{W}_1$ and assume that g is differentiable at x and f is differentiable at $g(x) \in \mathcal{W}_2$. Then the Fréchet differential of DF at x is the (composed) linear application $Df(g(x)) \circ Dg(x)$ that acts on any input v as :

$$DF(x)v = Df(g(x))(Dg(x)v). \quad (3.41)$$

Advanced Topics 3.29. Let $L : H \rightarrow K$ be a bounded linear operator between Hilbert spaces. The **transposed** (also called **adjoint**) operator $L^T : K \rightarrow H$ is the unique operator satisfying $\langle Lx, y \rangle_K = \langle x, L^T y \rangle_H, \forall x \in H, \forall y \in K$.

²When the function has several arguments one may also write $\nabla_y \mathcal{F}(x, y)$ to identify the argument that has been differentiated.

This is useful because for $\mathcal{F} : H \rightarrow \mathbb{R}$ ($H = \text{Hilbert space}$):

$$\nabla \mathcal{F}(x) = D\mathcal{F}^T(x)(1). \quad (3.42)$$

Note that in particular (3.41) can be written $D(f \circ g)^T = Dg^T \circ Df^T$. This means that to compute (or apply to some argument) $D(f \circ g)^T$ we use first Df^T and then Dg^T . This explains the **backward nature** of the gradient computation.

Example 3.30. We take now a matrix example: with previous notations denote $\mathcal{W}_1 := \mathbb{R}^{n_2 \times n_1}$ the space of matrices of size $n_2 \times n_1$, $\mathcal{W}_2 := \mathbb{R}^{n_2}$ and $g : \mathbb{R}^{n_2 \times n_1} \rightarrow \mathbb{R}^{n_2}$ defined by $g(W) = WY + b$ (matrix product between W and Y) with $Y \in \mathbb{R}^{n_1 \times 1}$, $b \in \mathbb{R}^{n_2}$ and $\mathcal{O} : \mathbb{R}^{n_2} \rightarrow \mathbb{R}$ some output function. We want to compute $\nabla_W(\mathcal{O} \circ g)$ using the gradient $\nabla \mathcal{O}$ of $\mathcal{O}$ at the point $g(W)$ written as **column** vector.

The space of matrices where W belongs is a Hilbert space endowed with the scalar product $\langle W_1, W_2 \rangle = \text{Tr}(W_1^T W_2)$ where $\text{Tr}(\cdot)$ is the usual trace operator. Take a variation w ; since g is affine, $g(W + w) = g(W) + wY$ so $Dg(w) = wY$. Moreover:

$$\begin{aligned} \mathcal{O}(g(W + w)) &= \mathcal{O}(g(W)) + \langle \nabla \mathcal{O}, g(W + w) - g(W) \rangle + o(\dots) \\ &= \mathcal{O}(g(W)) + \langle \nabla \mathcal{O}, Dg(w) \rangle + o(\dots) = \mathcal{O}(g(W)) + \langle \nabla \mathcal{O}, wY \rangle + o(\dots) \end{aligned} \quad (3.43)$$

Therefore $D(\mathcal{O} \circ g)(W)$ is the linear application $w \mapsto \langle \nabla \mathcal{O}, wY \rangle$. Furthermore, since:

$$\langle \nabla \mathcal{O}, wY \rangle = \text{Tr}(\nabla \mathcal{O}^T wY) = \text{Tr}(Y \nabla \mathcal{O}^T w) = \langle \nabla \mathcal{O} Y^T, w \rangle, \quad (3.44)$$

it follows that:

$$\nabla_W \mathcal{O}(WY + b) = \nabla \mathcal{O} Y^T \in \mathbb{R}^{n_2 \times n_1}. \quad (3.45)$$

Same kind of reasoning allows to write:

$$\nabla_Y \mathcal{O}(WY + b) = W^T \nabla \mathcal{O} \in \mathbb{R}^{n_1 \times 1}, \quad (3.46)$$

$$\nabla_b \mathcal{O}(WY + b) = \nabla \mathcal{O} \in \mathbb{R}^{n_2 \times 1}. \quad (3.47)$$